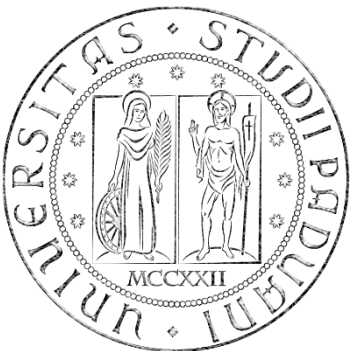


Biomedical Wearable Technologies
for Healthcare and Wellbeing

Implementing State Management

A.Y. 2025-2026

Giacomo Cappon



Outline

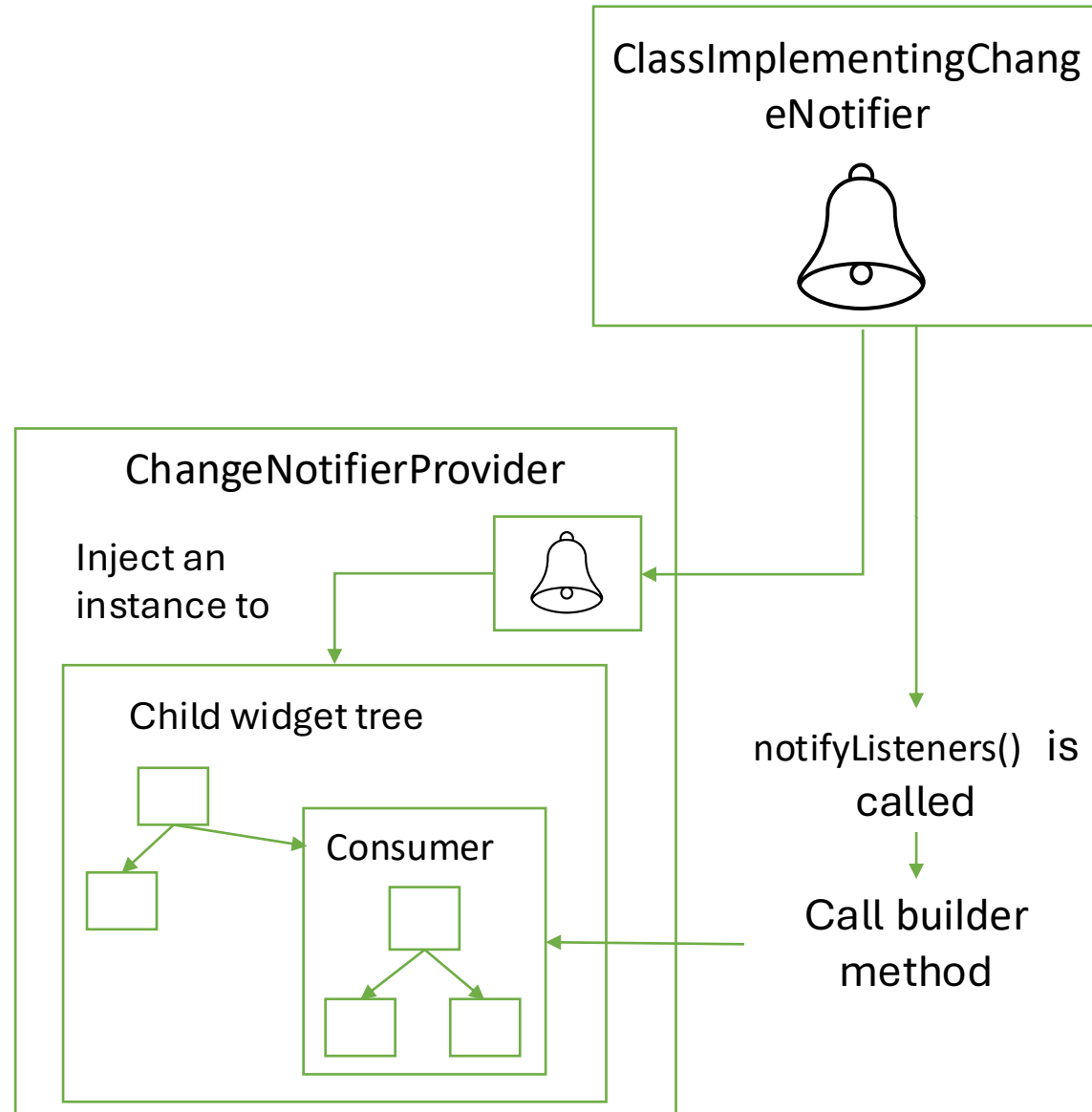
- **Recap on Provider and Case study**

- Exercises

- Homework

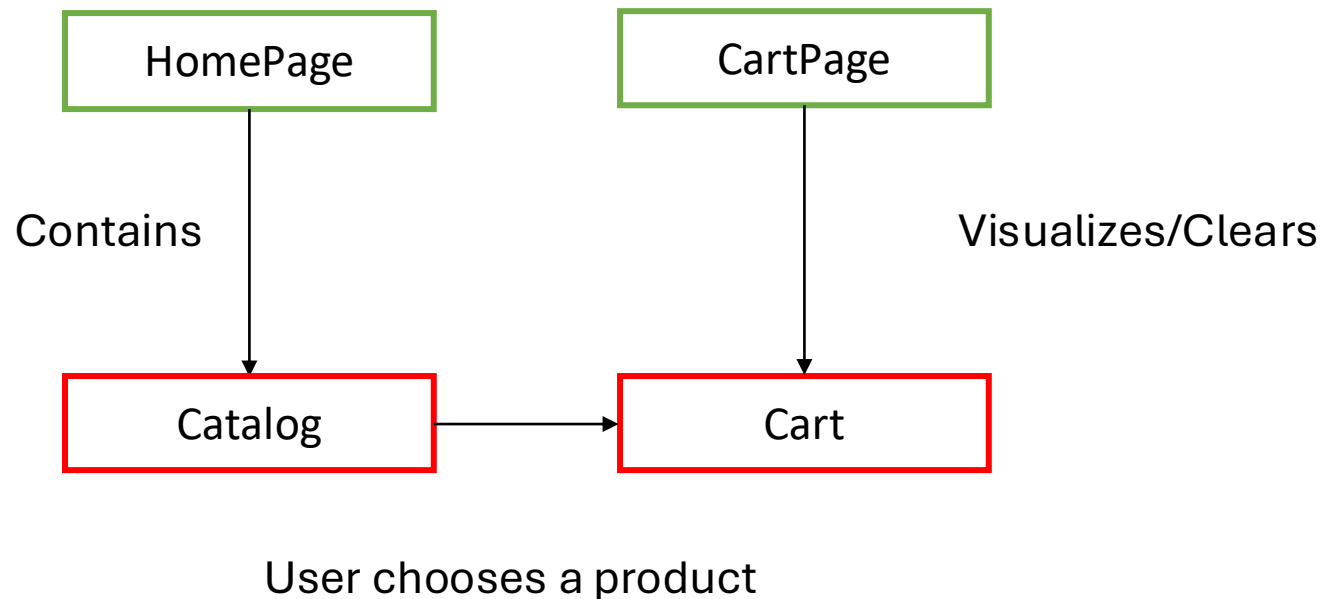
- Resources

Recap: Provider (core) flow

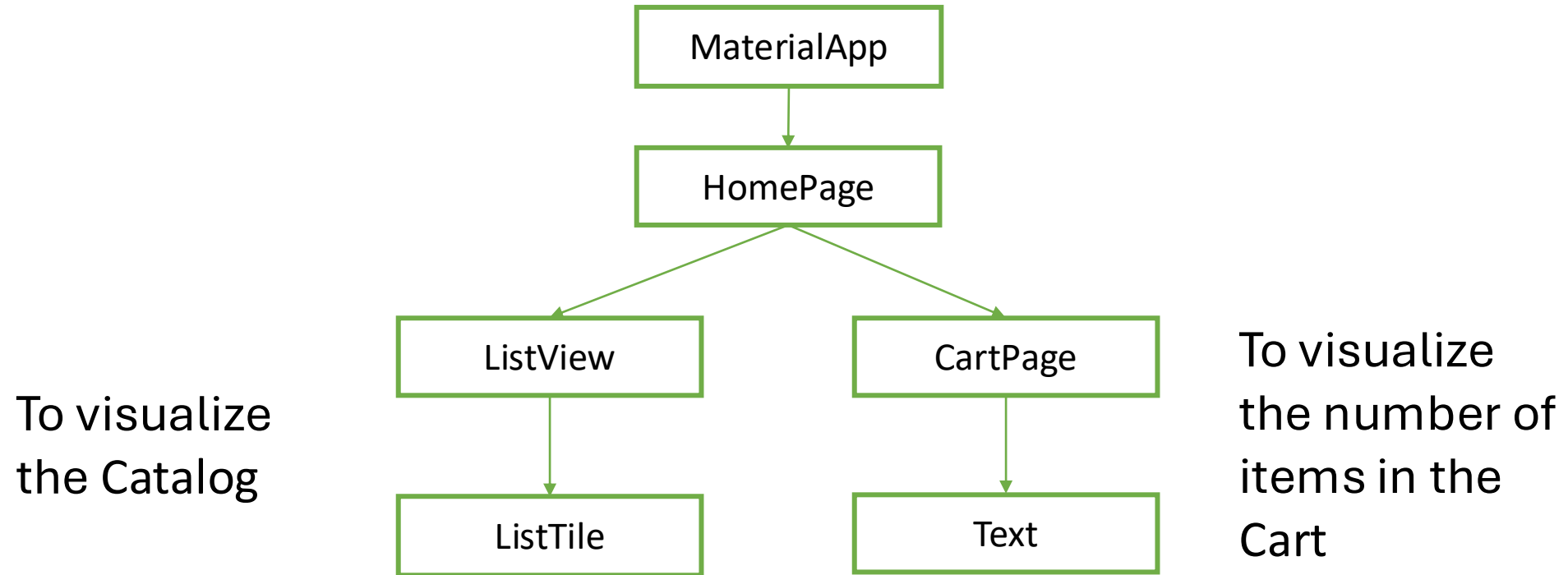


Case of study

- We will build a (too) simple e-commerce app. Users can choose, in the `HomePage`, products from a catalog and add them to the cart. In a different screen, `CartPage`, the number of items in the cart is visualized.



Case of study – UI widget tree



➤ How to manage the catalog and the cart? With two class (model).

- No problems with the Catalog (Ephemeral state)...
- On the other hand, Cart must be shared (App state) Where to put it?

Catalog

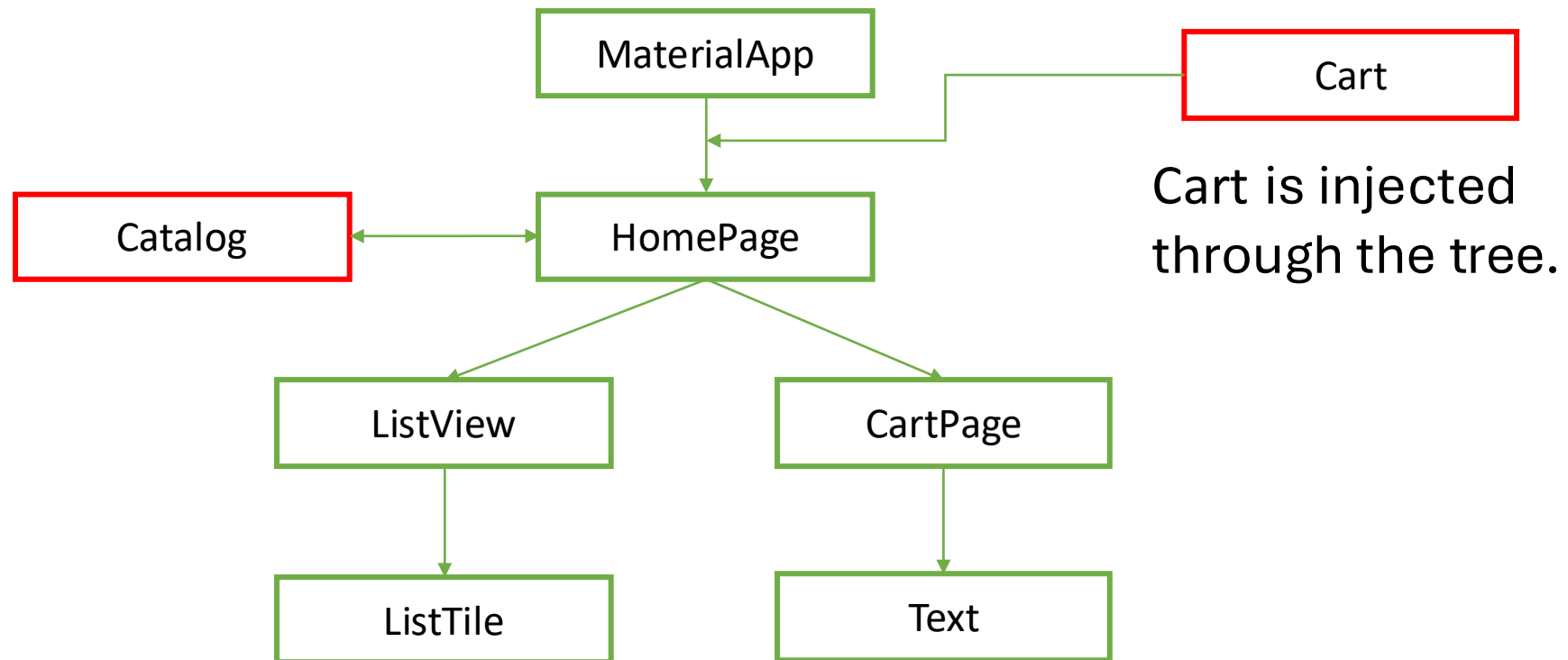
Cart

Lift the state up

- The idea is “Lift the state up”

$$\text{UI} = f(\text{state})$$

The layout on the screen Your build methods The application state



Catalog is just a variable of HomePage.

Case of study - Live

- Let's create a new project “shopper” that implements the case of study and shows how to use Provider.

```
lib
├── models
│   ├── cart.dart
│   └── catalog.dart
├── screens
│   ├── cartpage.dart
│   ├── homepage.dart
│   └── main.dart
```



Outline

- Recap on Provider and Case study
- **Exercises**
- Homework
- Resources

Exercise

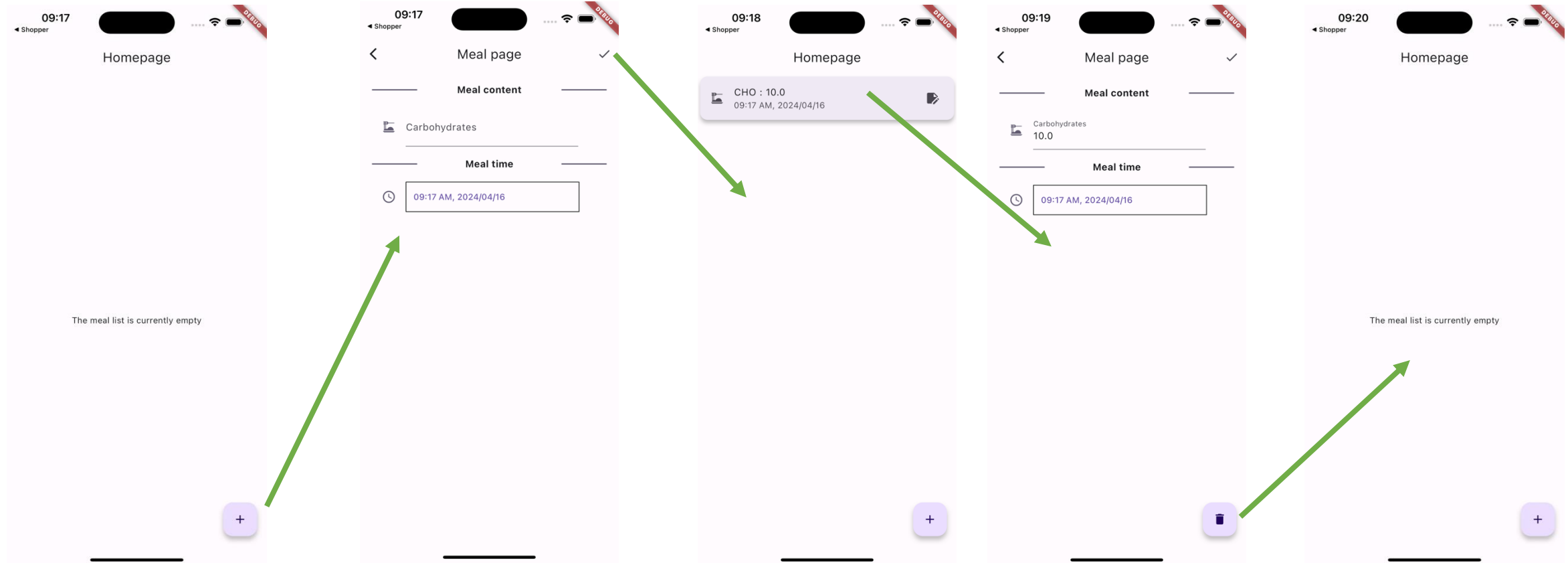
- Create a new project 'meal_manager', an app that stores the meal intakes of a user in terms of carbohydrate content and meal timing.

- The app needs to implement the following functionalities:
 - When a user opens (or restarts) the app, an empty list is showed;
 - By tapping a button, the user navigates to another page where he can add a new meal (CHO content and timing). Once he/she is done, the user taps a button and navigates back to the home page. The home page must show the updated meal list with the new meal just created;
 - (Bonus) The user can select a meal from the list. If he/she does that, he/she navigates to another page where he/she can edit or delete the meal entry from the list.

Exercise – Suggested steps

1. Create the UI with placeholders
2. Create a class that extends `ChangeNotifier`
3. "Give" that class to `ChangeNotifierProvider`
4. Implement the methods of the class created at step 2 (those that call `notifyListeners()`)
5. Implement the part of the code that "calls" the methods you implemented at step 4
6. Integrate the Consumer where you need them

Exercise: here's a possible idea



Homepage is initially empty

When the button is tapped the user navigates to another page where he/she can add a new meal

When the user finishes, the new meal is added to the list

When the meal is tapped, the user navigates to a new screen where he/she can edit or delete it

If the meal is edited/deleted the list is updated.

Exercise

- Create a new project “dynamic_listview”
- Reproduce the UI in the screenshot
- The FloatingActionButton on the bottom right has the following behaviour: whenever it is pushed, a new ListTile is added to the ListView and displayed to the user.



Exercise

- Hint: ListView has not just 1 but 4 constructors! ListView.builder could be of help:

```
List<String> items = ["1", "2", "3", "a", "b", "c"];  
ListView.builder(  
    itemCount: items.length,  
    itemBuilder: (context, index) {  
        return ListTile(title: Text(items[index]));  
    },  
)
```

- Note also that it is normal that when the app is reloaded or restarted, the ListView will look empty.



Outline

- Recap on Provider and Case study
- Exercises
- **Homework**
- Resources

Homework

- Get familiar with Provider
- Take a look to other approaches, e.g., Riverpod or BLoC. Maybe there is something that fits better your way of thinking!
- Take a look at my solution. You will find some useful code.

Outline

- Recap on Provider and Case study
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- **Resources**

Resources

➤ State management

- <https://docs.flutter.dev/development/data-and-backend/state-mgmt/intro>

➤ Making sense of all those Flutter Providers

- <https://medium.com/flutter-community/making-sense-all-of-those-flutter-providers-e842e18f45dd>

➤ List of state management approaches

- <https://docs.flutter.dev/development/data-and-backend/state-mgmt/options>